

An infinite-dimensional unital algebra has no separable singleton model

MathlibAnnex.Analysis.CStarAlgebra.Representation.not_singleton_amongNonUnital_of_infiniteDimensional

theorem

Uses the finite-dimensional algebra conclusion to contradict the proposed singleton condition.

Statement

Let A be an infinite-dimensional nonzero unital complex C^* -algebra, let H be a separable complex Hilbert space, and let $\pi : A \rightarrow \mathcal{B}(H)$ be a unital $*$ -representation. Then π cannot satisfy the singleton condition of R11.

Assumptions

Infinite dimensionality is a hypothesis on A . The conclusion excludes the conjunction of nonzero irreducibility and unitary equivalence with every nonzero irreducible $*$ -representation of A .

Conclusion

The specified π cannot represent a unique irreducible class. This does not rule out separable irreducible representations in general or construct a particular competing representation.

Proof route

Assume the singleton condition and apply the preceding theorem for the dimension of the algebra.

▼ Proof steps

1. If this π satisfied the proposed condition, then Finite dimension of the algebra from an unrestricted singleton model would apply: A is nonzero and unital, H is separable, and the assumed condition supplies its remaining input. It would give $\dim_{\mathbb{C}} A < \infty$, contrary to the stated infinite dimensionality. Therefore that condition cannot hold.

Main citations

- Exact theorem and proof
- The finite-dimensional algebra conclusion

Lean source signature (exact)

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theorem not_singleton_amongNonUnital_of_infiniteDimensional
  [Nontrivial A] (hA : ¬ FiniteDimensional ℂ A)
  [TopologicalSpace.SeparableSpace H]
  (pi : Representation A H) :
  ¬ IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi
```

In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
(hA : ¬ FiniteDimensional ℂ A)	A is not finite dimensional as a complex vector space.
[TopologicalSpace.SeparableSpace H]	The chosen Hilbert space H is separable.
(pi : Representation A H)	The specified unital $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$.
¬ IsSingletonIrreducibleModelAmongNonUnital.{u, v, u} pi	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.Representation.not_singleton_amongNonUnital_of_infiniteDimensional

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